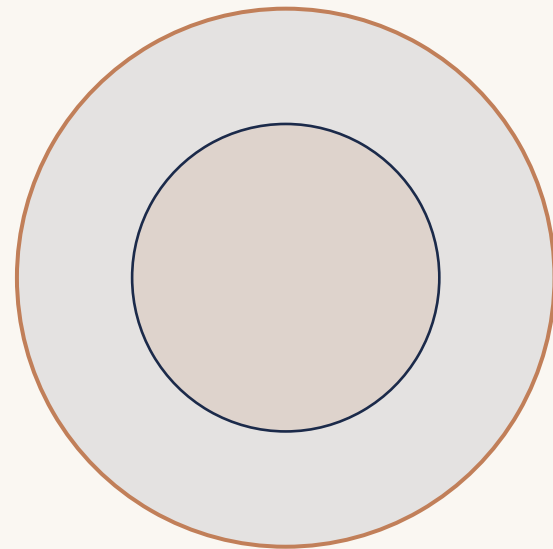


Helix Molecular

Accelerating drug discovery through
computational molecular intelligence.



Series A · Confidential · 2025

\$2.6B

Average cost to bring
a single drug to market

12 years

Average timeline from
discovery to approval

90%

Failure rate in clinical trials

The Problem

Drug discovery is broken.

Pharmaceutical companies spend billions screening millions of compounds, yet the vast majority of drug candidates fail in late-stage clinical trials. The fundamental bottleneck is molecular simulation — predicting how a candidate molecule will bind to a target protein with sufficient accuracy to avoid costly failures.

Current computational approaches rely on physics-based molecular dynamics simulations that take weeks to run for a single protein-ligand pair. This means researchers can only evaluate a tiny fraction of chemical space before committing to expensive wet-lab validation.

The result: promising compounds are missed, resources are wasted on poor candidates, and patients wait decades for treatments that could have been found faster.

Our Approach

Computational molecular intelligence that outperforms physics-based methods.

Helix Molecular combines graph neural networks with a proprietary molecular representation layer trained on 840 million protein-ligand interaction records. Our platform predicts binding affinity, toxicity, and ADMET properties in seconds rather than weeks.

Binding Affinity Prediction

Our model achieves a Pearson correlation of 0.89 on the PDBbind v2020 core set for binding affinity prediction, compared to 0.82 for the current state-of-the-art reported by Uni-Mol in 2023.

Simulation Speed

Helix screens 100,000 compounds against a target protein in under 4 hours on a single A100 GPU — a task that takes traditional molecular dynamics simulations approximately 6 months of compute time.

Toxicity Prediction

Achieves 97.2% AUC-ROC on the Tox21 dataset, surpassing all published methods including Chemprop and MolBERT by over 8 percentage points.

Pipeline & Traction

From prediction to the clinic.

Helix is the first AI drug discovery platform to advance a computationally-designed small molecule into Phase I clinical trials without any prior wet-lab lead optimization.

3

Pharma Partners

Active collaborations with two top-20 pharma companies and one mid-cap biotech

1

Phase I Candidate

HLX-0042, a novel kinase inhibitor for NSCLC, entered Phase I in Q3 2024

840M

Training Interactions

Proprietary dataset built from public and licensed sources over 3 years

\$4.2M

ARR

Platform licensing revenue from pharma discovery partners

Our world-class team of computational chemists and machine learning researchers is pioneering the future of precision medicine.

Helix Molecular

Raising \$18M Series A to bring computational
molecular intelligence to every drug program.

founders@helixmolecular.com

helixmolecular.com